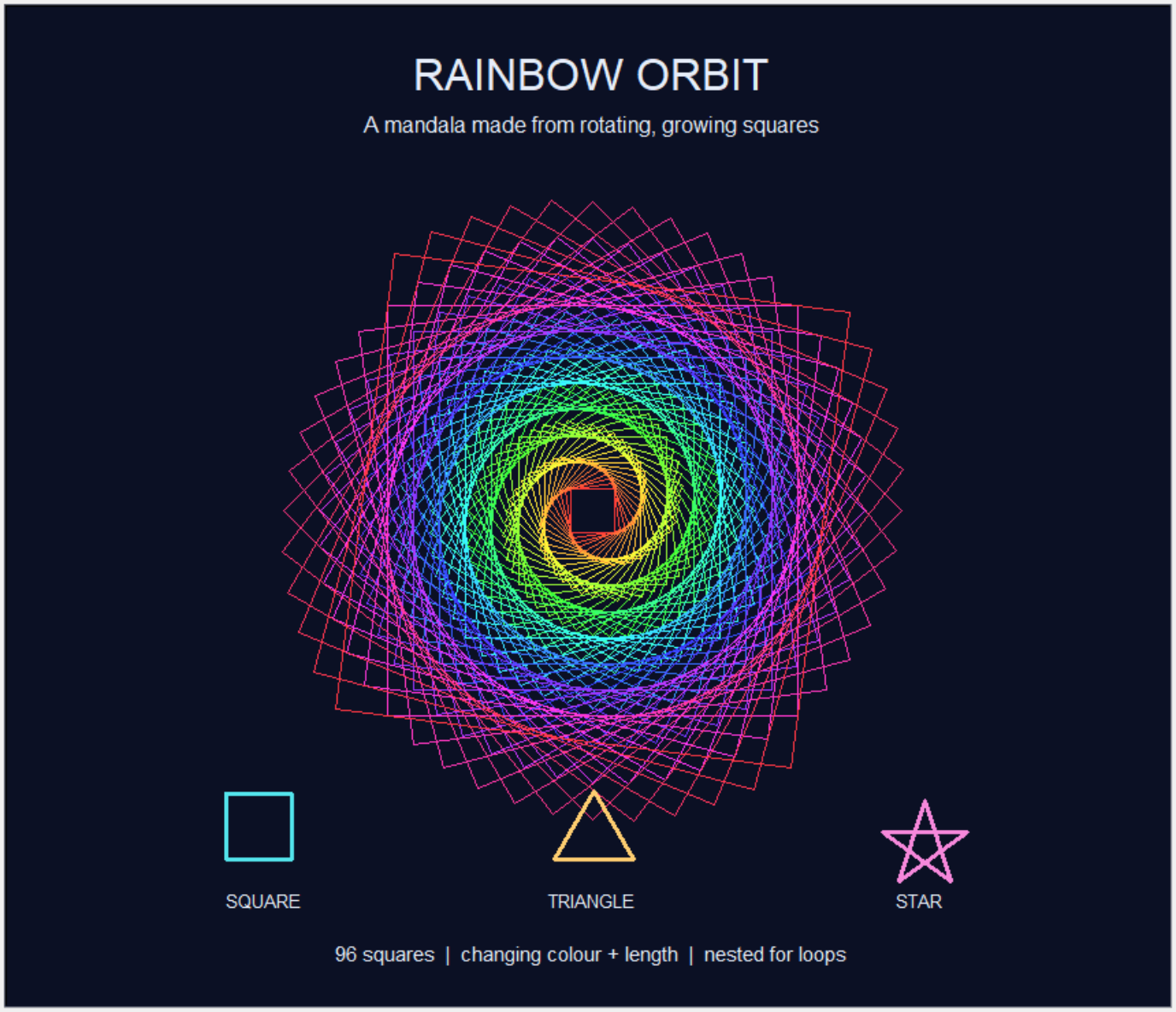
Assignment One

Name: Fan Yi

GitHub:
github.com/15895656407fanyi-byte/ELEC7023_Assignment1_FanYi

Turtle Drawing



An outer for loop draws 96 squares, rotating each square by 7.5 degrees.
Each iteration changes the hue and increases the side length from 32 to 336 pixels.
An inner for loop draws four equal sides with a 90-degree turn after each side.
The overlapping squares form the rainbow pattern. Separate loops draw the
square, triangle and five-pointed star shown at the bottom.

QA Bot

```
PS C:\Users\FanYi\Desktop\2026S2\ELEC7023> python assignment1_FanYi.py

ASSIGNMENT ONE
1 - Turtle Drawing
2 - QA Bot
3 - Exit
Choose 1, 2 or 3: 1
Drawing complete. Close the drawing window to return to the menu.

ASSIGNMENT ONE
1 - Turtle Drawing
2 - QA Bot
3 - Exit
Choose 1, 2 or 3: 2

QUESTION ANSWERING BOT
Topics: HKU, robot, Jetson, AI, human. Type 'quit' to return.
For multiple keywords, priority is HKU > robot > Jetson > AI > human.
You: HKU
Bot: HKU stands for The University of Hong Kong.
You: robot
Bot: A robot is a programmable machine that can sense and act.
You: Jetson
Bot: NVIDIA Jetson is a computing platform for AI and robotics at the edge.
You: AI
Bot: AI means Artificial Intelligence: computers performing tasks such as recognition.
You: human
Bot: A human is a person; humans design, train and use robots and AI systems.
You: rain
Bot: Sorry, I do not know that topic. Try HKU, robot, Jetson, AI or human.
You:
Bot: Please type a question about HKU, robot, Jetson, AI or human.
You: quit
Bot: Goodbye!

ASSIGNMENT ONE
1 - Turtle Drawing
2 - QA Bot
3 - Exit
Choose 1, 2 or 3: 3
Thank you. Goodbye!
PS C:\Users\FanYi\Desktop\2026S2\ELEC7023>
```

The bot repeatedly reads questions with input() until the user enters quit, exit or bye.

Questions are converted to lowercase and split into words for keyword matching.

An if / elif / else chain selects a response for HKU, robot, Jetson, AI or human.

If several keywords appear, the first matching branch determines the answer.

Empty input prompts for a question; an unknown topic receives a fallback reply.